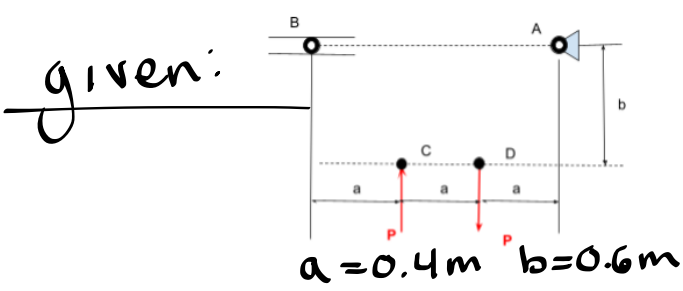


Assignment 2 Design truss system



A is a pin joint
B is a roller joint
 $P = 20\text{kN} \leftrightarrow 30\text{kN}$
chosen

20kN chosen

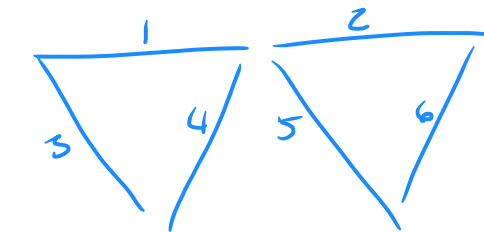
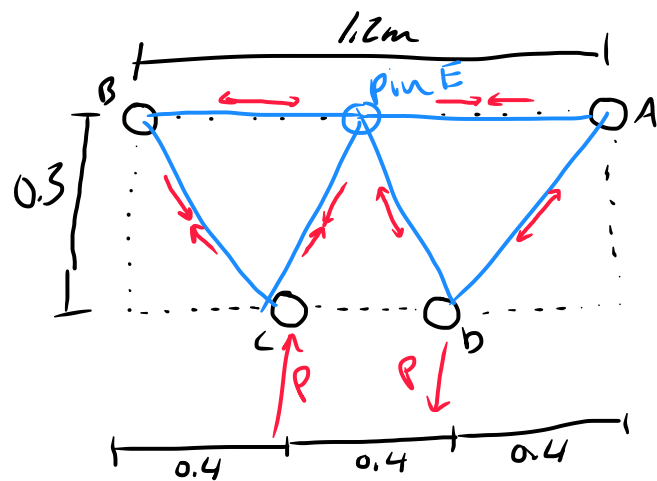
the cross sectional Area of each element has to be identical.

the pins have to be identical and each element has to have the same cross sectional geometry.

Material: A500 Structural steel or similar

Step 1: Sketch truss structure for design and lengths.

need to support loads at C and D



Lengths

truss	meters
1	0.6
2	0.6
3	0.5
4	0.36055
5	0.36055
6	0.5

Symmetry

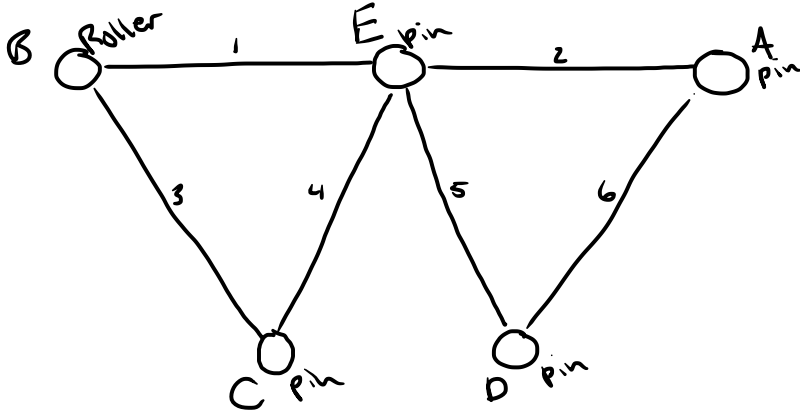
$$3 = 6$$

$$4 = 5$$

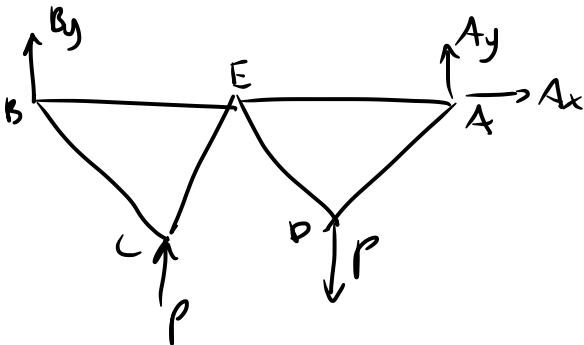
$$3 = 6 = \sqrt{0.3^2 + 0.4^2} = 0.5$$

$$4 = 5 = \sqrt{0.3^2 + 0.2^2} = 0.36055$$

Step 2: A good FBD of the trusses



Step 3: Symbolically solve internal forces



$$\sum F_x = 0 = A_x$$

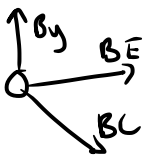
$$\sum F_y = A_y + B_y + P - P = 0$$

$$\sum F_y = A_y + B_y = 0$$

$$A_y = -B_y$$

$$\sum M_A = 0.4(P) - 0.8(P) - 1.2(B_y) = 0$$

pin B



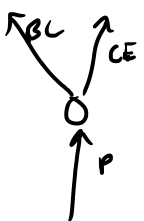
$$\sum F_y = B_y - BC_y = 0$$

$$BC_y = B_y$$

$$\sum F_x = BE + BC_x = 0$$

$$BE = -BC_x$$

pin C

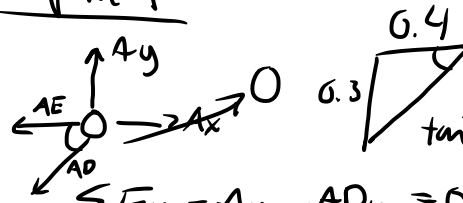


$$\sum F_y = BC_y + CE_y + P = 0$$

$$\sum F_x = CE_x - BC_x = 0$$

$$CE_x = BC_x$$

pin A



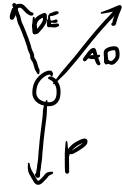
$$\sum F_y = A_y - AD_y = 0$$

$$A_y = AD_y$$

$$\sum F_x = -AE_x - AD_x = 0$$

$$AD_x = -AE_x$$

pin D



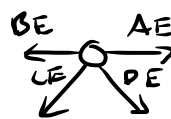
$$\sum F_y = AD_y - P + DE_y = 0$$

$$DE_y = P - AD_y$$

$$\sum F_x = AD_x - DE_x = 0$$

$$DE_x = AD_x$$

pin E



$$\sum F_x = AE + DE_x - CE_x - BE = 0$$

$$\sum F_y = -DE_y - CE_y = 0$$

$$CE_y = -DE_y$$

Step 4: Numerically solve inter

$$\sum M_A = 0.4(P) - 0.8(P) - 1.2(B_y) = 0$$

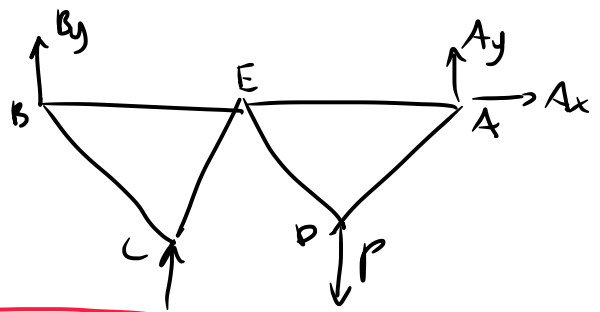
$$1.2(B_y) = -0.4(P)$$

$$P = 20 \text{ kN}$$

$$B_y = \frac{-0.4P}{1.2}$$

$$= \frac{-0.4 \cdot 20}{1.2} \text{ kN} = -6.6 \text{ kN } B_y$$

$$A_y = -B_y = 6.6 \text{ kN } A_y \quad \sum F_x = 0 = A_x$$



$$\theta = 36.87^\circ$$

$$A_y = AD_y = 6.6 \text{ kN}$$

$$\sin \theta = \frac{6.6}{AD} \quad \tan \theta = \frac{6.6}{AD_x}$$

$$AD = \frac{6.6}{\sin \theta} = 11.1 \text{ kN } AD$$

$$AD_x = \frac{6.6}{\tan \theta} = 8.8 \text{ kN}$$

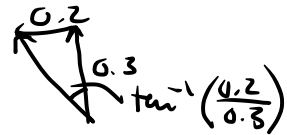
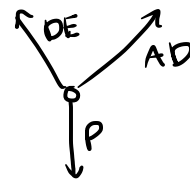
$$AD_x = -AE = -8.8 \text{ kN } AE$$

$$DE_y = P - AD_y$$

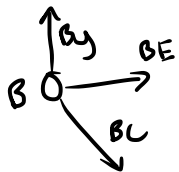
$$DE_y = 20 - 6.6 = 13.3 \text{ kN}$$

$$\cos \phi = \frac{13.3}{DE}$$

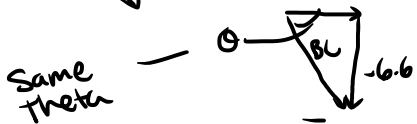
$$DE = \frac{13.3}{\cos \phi} = 16.025 \text{ kN}$$



$$\phi = 33.69^\circ$$



$$B_y = BC_y = -6.6$$



$$\sin \theta = \frac{-6.6}{BC}$$

$$BC_x = -BE$$

$$BC_x = \frac{-6.6}{\tan \theta} = -8.8 \rightarrow BE = 8.8 \text{ kN}$$

$$BC = \frac{-6.6}{\sin \theta} = -11.1$$

EXTERNAL FORCES

$$A_x = 0$$

$$A_y = 6.6 \text{ kN}$$

$$B_y = -6.6 \text{ kN}$$

$$P_1 = P_2 = 20 \text{ kN}$$

INTERNAL FORCES

$$AD = 11.1 \text{ kN}$$

T

$$AE = 8.8 \text{ kN}$$

C

$$BC = 11.1 \text{ kN}$$

C

$$BE = 8.8 \text{ kN}$$

T

$$CE = 16.025 \text{ kN}$$

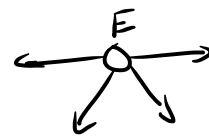
C

$$DE = 16.025 \text{ kN}$$

T

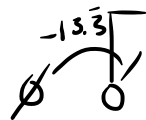
T: Tension

C: Compression



$$CE_y = -DE_y$$

$$CE_y = -13.3 \text{ kN}$$



$$\cos \phi = \frac{-13.3}{CE}$$

$$CE = \frac{-13.3}{\cos \phi} = -16.025 \text{ kN } CE$$

Step 5: Cross sectional Area of truss members

knowns:

largest internal force F_{max}
Material
Safety factor

unknowns:

Stress
material yield strength
Cross sectional Area
needed with Safety factor
Density

$F_{max} = 16025 \text{ N}$ tension & compression

Material $\rightarrow$ yield strength

A500 Structural Steel $\rightarrow$ can be many numbers
 $\rightarrow$ assuming 250 MPa Megapascals

Stress is defined as: $\sigma = \frac{F}{A}$

250 MPa is a measurement of stress

to find how much area needed
we need to rearrange and redefine the stress equation

$\sigma \rightarrow \sigma_y$
yield strength

$F \rightarrow F_{max}$

then the Safety factor (3.5)

$$A = \frac{F_{max}}{\sigma_y}$$

$$3.5 \cdot A = \frac{3.5 \cdot F_{max}}{\sigma_y}$$

$$\text{Area needed} = \frac{3.5 F_{max}}{\sigma_y}$$

plugging numbers in:

$$A = \frac{3.5 (16025)}{250 \cdot 10^6} \frac{\text{N}}{\text{m}^2}$$

$$A = 6.0022435 \text{ m}^2 \cdot \frac{(1000 \text{ mm})^2}{1 \text{ m}^2}$$

$$224.435 \text{ mm}^2$$

$$2.2435 \text{ cm}^2$$

Total length of All members (cm)

$$60 + 60 + 50 + 50 + 36055 + 36055 = 242.11 \text{ cm}$$

$$L \cdot A = \text{Volume} = 655.3488 \text{ cm}^3$$

$$\text{Density} = 5111.70523 \text{ grams}$$

$$7.8 \text{ g/cm}^3 \rightarrow 5.112 \text{ kg}$$

Step 4: Numerically solve inter

$$\sum M_A = 0.4(P) - 0.8(P) - 1.2(B_y) = 0$$

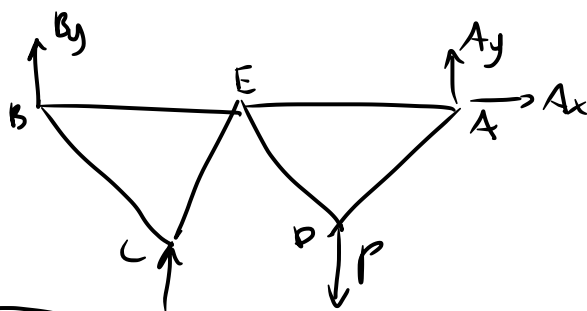
$$1.2(B_y) = -0.4(P)$$

$$P = 20 \text{ kN}$$

$$B_y = \frac{-0.4P}{1.2}$$

$$= \frac{-0.4 \cdot 20}{1.2} \text{ kN} = -6.6 \text{ kN } B_y$$

$$A_y = -B_y = 6.6 \text{ kN } A_y \quad \sum F_x = 0 = A_x$$

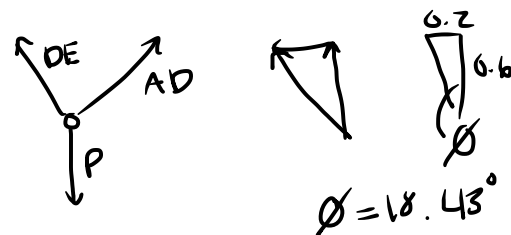


$$\theta = 56.3^\circ$$

$$A_y = AD_y = 6.6 \text{ kN}$$

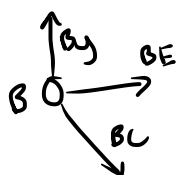
$$\sin \theta = \frac{6.6}{AD} \quad \tan \theta = \frac{6.6}{AD_x} \quad AD_x = \frac{6.6}{\tan \theta} = 4.4$$

$$AD = \frac{6.6}{\sin \theta} = 8.0123 \text{ kN } AD \quad AD_x = -AE \quad AE = -4.4 \text{ kN}$$

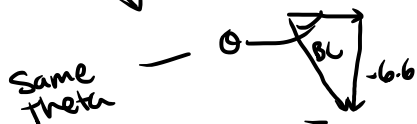


$$DE_y = P - AD_y = 20 - 6.6 = 13.3 \text{ kN}$$

$$\cos \phi = \frac{13.3}{DE} \quad DE = 14.055 \text{ kN}$$



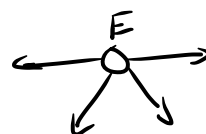
$$B_y = B_c = -6.6$$



$$\sin \theta = \frac{6.6}{BC} \quad BC_x = \frac{-6.6}{\tan \theta} = -4.4$$

$$BC = \frac{-6.6}{\sin \theta} = -8.0123 \text{ kN } BC$$

$$BC_x = -BE \quad BE = 4.4 \text{ kN}$$



$$CE_y = -DE_y$$

$$CE_y = -13.3 \text{ kN}$$

$$\cos \phi = \frac{13.3}{CE} \quad CE = 14.055 \text{ kN}$$

EXTERNAL FORCES

$$A_x = 0$$

$$A_y = 6.6 \text{ kN}$$

$$B_y = -6.6 \text{ kN}$$

$$P_1 = P_2 = 20 \text{ kN}$$

INTERNAL FORCES

$$\begin{aligned} AD & 8.0123 \text{ kN } T \\ AE & 4.4 \text{ kN } C \\ BC & 8.0123 \text{ kN } C \\ BE & 4.4 \text{ kN } T \\ CE & 14.055 \text{ kN } T \\ DE & 14.055 \text{ kN } C \end{aligned}$$

T: Tension
C: Compression

Step 5: Cross sectional Area of truss members

knowns:

largest internal force F_{max}
Material
Safety factor

unknowns:

Stress
material yield strength
Cross sectional Area
needed with Safety factor
Density

$$F_{max} = 14,055 \text{ N} \text{ tension \& compression}$$

Material $\rightarrow$ yield strength

A500 Structural Steel $\rightarrow$ can be many numbers
 $\rightarrow$ assuming 250 MPa Megapascals

Stress is defined as: $\sigma = \frac{F}{A}$

250 MPa is a measurement of stress

to find how much area needed
we need to rearrange and redefine the stress equation

$\sigma \rightarrow \sigma_y$
yield strength

$F \rightarrow F_{max}$

then the Safety factor (3.5)

$$A = \frac{F_{max}}{\sigma_y}$$

$$3.5 \cdot A = \frac{3.5 \cdot F_{max}}{\sigma_y}$$

$$\text{Area needed} = \frac{3.5 F_{max}}{\sigma_y}$$

plugging numbers in:

$$A = \frac{3.5 (14055)}{250 \cdot 10^6} \frac{\text{N}}{(\text{N/m}^2)}$$

$$A = 0.0019677 \text{ m}^2 \cdot \frac{(100 \text{ cm})^2}{(1,0000 \text{ m}^2)}$$
$$1.9677 \text{ cm}^2$$

Total length of All members (cm)

$$60 + 60 + 72.111 + 72.111 + 63.246 + 63.246 = 346.714 \text{ cm}$$

$$L \cdot A = \text{Volume} = 749.13 \cdot \text{Density} = 5843.22 \text{ grams}$$

$$7.8 \text{ g/cm}^3 \rightarrow 5.843 \text{ kg}$$

Step 6: Pin Shear

knowns:

internal forces

pins are hardened tool steel

yield Shear Strength: 170 ksi

pin density 0.278 lb/in³

unknowns:

Pin Area

$$\begin{aligned} DE &= 16.625 \text{ kN} \\ AD &= 11.7 \text{ kN} \\ P &= 20 \text{ kN} \end{aligned}$$

previously found

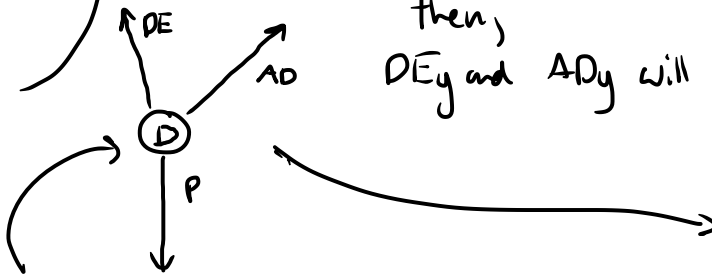
$$\begin{aligned} AD_y &= 6.6 \text{ kN} \\ DE_y &= 13.3 \text{ kN} \end{aligned}$$

DE_x and AD_x cancel

then,

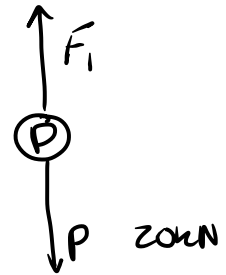
DE_y and AD_y will equal P

$$\bar{F}_1 = DE_y + AD_y = 20 \text{ kN}$$



D has the highest forces applied because it's the closest to the load.

D is also a single shear connection because the load is not a connection



$$F_1 = P$$

$$\tau = \frac{F}{A} \rightarrow \tau_y = \frac{\bar{F}_1}{A_{pin}} \rightarrow A_{pin} = \frac{\bar{F}_1}{\tau_y} \text{ ksi} \rightarrow \text{MPa}$$

$$170 \text{ ksi} \cdot \frac{6.89476}{1 \text{ ksi}} = 1172.11 \text{ MPa}$$

$$A_{pin} = (SF) \frac{\bar{F}_1}{\tau_y}$$

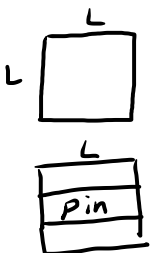
$$\frac{20,000 \text{ N}}{1172.11 \text{ N/mm}^2}$$

$$\frac{\text{N}}{\text{mm}^2}$$

$$= 17.06 \text{ mm}^2 \text{ without safety factor}$$

$$68.25 \text{ mm}^2 \text{ Cross sectional Area}$$

if the truss is a square,



$$\begin{aligned} L &= \sqrt{224.435} \text{ mm} \\ L &= 14.98 \text{ mm} \end{aligned}$$

Volume of the pin would be:

$$L \cdot A = 14.98 \text{ mm} \cdot 68.25 \text{ mm}^2 = 1022.385 \text{ mm}^3$$

density of the pins 0.278 lb/in³

$$0.278 \frac{\text{lb}}{\text{in}^3} \cdot 1022.385 \cdot \frac{\text{in}^3}{(25.4 \text{ mm})^3} = \frac{453.5429}{16}$$

$$\sqrt{\frac{68.25}{\pi}} \cdot 2 = 9.52 + 14.98 = 24.3 \text{ mm}$$

diameter of joint

$$= 7.867 \frac{\text{grams}}{\text{pin}} \cdot 5 \text{ pins} = 39.335 \text{ grams}$$